1. **5. Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation**
    - Core skill: reading numerical values from a visual representation (bar chart) and interpreting them to evaluate whether each statement matches the data.
2. **1. Interpret a mathematical result back into the real world context**
    - Relating values from the chart to real-world concepts like “average chicks per pair,” “percentage of pairs raising a chick,” and “extinction prediction,” then making true/false judgments.
3. **2. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem**
    - Needed especially for statement 3 (“extinct by about 2015”): judging whether such a prediction is reasonable given limited data on breeding success, recognizing the unreliability of extrapolating to extinction from a few years of data.
4. **9. Understand the relationship between the context of the problem and representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution**
    - Statement 3 highlights this—must understand that a chart of breeding success for only a few years cannot reliably support long-term extinction predictions; this involves gauging the feasibility and limitations of the inference.
5. **3. Identify the limits of the model used to solve a problem**
    - Again concerning statement 3: recognizing the limitations of using a short time series to infer long-term population survival.
6. **8. Recognise [demonstrate, interpret, explain] the extent and limits of mathematical concepts and mathematical solutions**
    - Interpreting statement 2: understanding the relationship between “average chicks per pair” and “percentage of pairs raising a chick”—if the average is x and each pair raises at most 1 chick, then at most $x \times 100\%$ of pairs raised a chick. This involves recognizing logical limits of the mathematical concept.
7. **7. Construct and communicate explanations and arguments in the context of the problem**
    - Although the task only requires circling true/false, the student must internally (or in reasoning) construct explanations for why each choice is made.

1. **5. Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation**
   - 最核心的是从柱状图这种视觉化数据格式中读取数值并解释成具体结论，评价每个陈述是否与图形数据一致。
2. **1. Interpret a mathematical result back into the real world context**
   - 将图表中的数值与“平均每对养育雏鸟数”“百分比”“灭绝预测”等现实情境相联系，做出真/假判断。
3. **2. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem**
   - 需要判断“到 2015 年会灭绝”这种推断是否合理：仅凭有限年份的繁殖数据推断灭绝，不符合模型外推的谨慎性，因此判断为 False 需要理解这种推断的不可靠性。
4. **9. Understand the relationship between the context of the problem and representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution**
   - 第 3 个陈述（关于灭绝）尤其体现这一点，必须明白图表只能给出部分年份繁殖成功率，不能直接推断灭绝，这是模型/数据的限制。
5. **3. Identify the limits of the model used to solve a problem**
   - 同样是针对第 3 个陈述，需认识到用几年的时间序列数据做长期种群生存推断的限制。
6. **8. Recognise [demonstrate, interpret, explain] the extent and limits of mathematical concepts and mathematical solutions**
   - 解释为什么第 2 个陈述的推理可能成立或不成立：图表中平均雏鸟数如果低于 0.8，是否意味着少于 80% 的夫妇养了雏鸟？这里涉及“平均雏鸟数”与“养雏夫妇比例”的数学关系理解，如果平均数为 x，最多 x% 的夫妇养了雏鸟（如果每对最多养 1 只），因此需要判断这种逻辑界限。
7. **7. Construct and communicate explanations and arguments in the context of the problem**
   - 虽然题目只要求判断真/假，但学生在心里或作答时需要给出内部推理，解释为何这样判断。